Physical Chemistry

November 30, 2007

1. What is a symmetry operation? (5 points)

2. What is a point group? (5 points)

3. Describe five kinds of symmetry operations that can form a point group. (5 points)

4. Identify the point groups (using the Schoenfliess notation) to which H_20, BF_3, and C_6H_6 belong, and list their most important symmetry elements. (10 points)

5. Using the 1s orbitals on each atom of NH_3 as a basis, construct the matrix representatives for the symmetry operations of point group C_{3v}. (25 points)

6. Determine the character of the matrix representation, Γ, of C_{3v} constructed in problem 5. (25 points)

7. Determine what irreducible representations are contained in the reducible representation Γ constructed in problem 5. (25 points)

Help: $a_\mu = (1/g) \sum_{i=1,k} [\ g_i\ \chi_i^{(\mu)}\ \chi_i^{(\Gamma)}\ ]$ where $\chi_i^{(\Gamma)}$ is the character, in the representation Γ, of an element belonging to one of the k classes K_i; $\chi_i^{(\mu)}$ is the character of the irreducible representation of the point group in class i; g_i is the number of elements of class i; g is the number of elements of the point group.

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0